Reproducibility conditions for open desktop robot arms

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Abstract

Open desktop robot arms offer accessible platforms for education and embodied intelligence, but a public repository does not establish that another laboratory can reproduce their behavior. This critical narrative review examines the conditions connecting mechanical design, procurement, assembly, commissioning, licensing, and performance evidence. Eight purposively selected project families span smart-servo chains, stepper-driven mechanisms, hybrid actuation, and antagonistic tendon systems. Publications and official project artifacts were inspected through 19 October 2026, with repository revisions and claim-level evidence retained. A seven-domain framework separates the availability of artifacts from demonstrated reconstruction and experimental replication. The comparison identifies consequential distinctions between editable designs and printable meshes, repository-level and component-level permissions, configuration procedures and performance tests, and nominal specifications and measurements with stated conditions. Published GummiArm load measurements illustrate why a payload value requires a pose and hardware version. Examples from robot learning further show why matching geometry alone does not establish equivalent observations, actions, or contact behavior. The review proposes a minimum release package linking build identity, calibration, test conditions, and maintenance records. These recommendations support auditable platform selection and reporting; they do not constitute a validated readiness score or evidence of successful reconstruction.

Keywords: open-source hardware; desktop robot arms; reproducibility; mechanical design; assembly documentation; embodied intelligence

1 Introduction

A desktop robot arm can be inexpensive to acquire yet costly to turn into a dependable research instrument. The practical difficulty extends beyond printing links or downloading a controller. A builder must connect a mechanical revision to obtainable parts, correct wiring, calibrated motion, and a comparable test. Open robot hardware reviews already identify documentation, fabrication access, and maintenance as central concerns [1]. Reproducible robotics likewise requires sufficient information to evaluate the experimental conditions, rather than treating source availability as the endpoint [2, 3].

This distinction matters when desktop arms support embodied intelligence. Demonstrations, contact interactions, and learned policies depend on the physical system that generates them. A change in joint zero, tendon tension, gearing, or camera attachment may alter the meaning of a recorded action. Operating one assembled device therefore answers a narrower question than reconstructing it elsewhere. The GummiArm project connects these issues directly through a printable platform designed for experiments involving variable stiffness and physical interaction [4]. Its relevance here is the connection between embodiment and experimental conditions, rather than an assumption that compliance guarantees reproducibility.

The present review addresses bench-mountable arms whose public releases invite construction, modification, or research use. “Desktop” denotes the installation context, not a standardized reach,

mass, or safety category. The scope includes larger bench-mounted research prototypes such as GummiArm, while excluding complete mobile and humanoid systems from the core comparison. Commercial actuators remain within scope because rebuilding an arm need not require manufacturing every motor. However, purchasing a component and possessing its design source represent different levels of openness.

Three neighboring literatures establish the need for a narrower synthesis. Patel and colleagues review open hardware across robotics and provide development practices [1]. Mustary and colleagues examine manipulator stability, including control, stiffness, and mechanical parameters [5]. Doloi and colleagues consider low-cost additive manufacturing for laboratory automation [6]. Their respective scopes inform this review, but the present comparison follows the dependencies between a desktop arm release and a reproducible experimental configuration. Table 1 distinguishes these questions without claiming that earlier work ignored reproducibility.

Table 1. Positioning relative to neighboring reviews.

Review	Primary scope	Relationship to this review
Patel et al. [1]	Open hardware across robotics and development practices	Foundation for documentation and fabrication concerns; this review adds configuration-specific desktop-arm comparisons.
Mustary et al. [5]	Manipulator stability, control, stiffness, and mechanical parameters	Explains performance-sensitive parameters; this review follows the public evidence needed to reconstruct and test them.
Doloi et al. [6]	Low-cost additive manufacturing for laboratory automation	Related application and fabrication scope; abstract and metadata only were used here.
Present review	Eight desktop-arm project families and their reproduction conditions	Connects design, procurement, assembly, rights, calibration, performance evidence, and change records.

The review asks three questions. Which public artifacts support construction and operation of a specified arm? How do mechanical architectures change the calibration and reporting information required? What evidence permits a performance claim to travel from one device or laboratory to another? These questions lead to three contributions: a seven-domain framework, a version-linked comparison of eight project families, and a minimum release package for desktop manipulation research. The framework is an organizing proposal grounded in the inspected sources, not an experimentally validated predictor of build success.

The term “open-source” also requires care. The Open Source Hardware Association (OSHWA) definition concerns usable design sources and rights to study, modify, distribute, make, and sell hardware [7]. A noncommercial release can provide valuable construction information while failing that definition. For this reason, the comparison retains restrictive and partially open projects as explicit boundary cases. It does not certify any project or infer permissions for purchased components.

2 Scope and evidence method

2.1 Review design and search

A critical narrative review was chosen because the evidence includes technical papers, repository files, licenses, bills of materials (BOMs), and assembly instructions. These sources answer different questions and do not support a common effect estimate. Narrative-review quality guidance emphasizes a clear rationale, transparent literature searching, appropriate referencing, and reasoned

scientific argument [8]. Search reporting was also informed by PRISMA-S, without representing this review as a systematic review or claiming PRISMA compliance [9].

Searches were executed on 19 October 2026 using the available web-search service, publisher sites, Crossref DOI lookups, and official project repositories. The search terms combined “open source,” “robot arm,” “desktop,” “low cost,” “review,” “reproducibility,” “GummiArm,” “OpenMANIPULATOR,” and named projects. Citation and project-link following supplemented these searches. No Scopus, Web of Science, or subscription-database search was performed. Supporting Information S1 records executed queries, source routes, access failures, and the limits of the search record.

The core sample was selected purposively for variation in architecture and release practice. GummiArm provided the starting case. Thor, BCN3D Moveo, OpenMANIPULATOR-X, Niryo One, Poppy Ergo Jr, PAROL6, and SO-101 supplied contrasting fabrication, actuation, documentation, and licensing arrangements. Selection required a bench-mountable articulated arm and inspectable official hardware-related artifacts. Neither a peer-reviewed hardware paper nor unrestricted licensing was required. This choice avoids equating publication status with build documentation, but it also limits representativeness.

These eight families are cases for analytical comparison, not the outcome of exhaustive screening. Other arms, including PARA and contemporary larger research platforms, may fit related scopes. They were not coded in the core matrix. ALOHA and GELLO are considered only as contextual evidence about teleoperation and learning-system reproduction. ROBEL provides a comparator for explicit replication reporting outside the desktop-arm category. No reconstructed screening totals, duplicate-removal counts, or study-selection flow statistics are reported.

2.2 Source inspection and extraction

For each project, extraction addressed design representation, purchased components, assembly guidance, commissioning, licensing, performance evidence, and version relationships. Official sources took precedence over third-party catalogs. Scientific claims were checked against accessible article text or author-hosted full text. Bibliographic fields were checked against publisher, proceedings, or Crossref records. The Doloi review was used only for scope positioning from its accessible abstract and metadata, because its full text could not be retrieved.

Repository snapshots retain commit identifiers, file inventories, and downloaded evidence files. Linked documentation was inspected where accessible. A repository tree establishes that named files are present. It does not establish that a CAD assembly opens correctly or that every part fits. Videos and cloud-hosted CAD were not treated as inspected merely because a link existed. For example, Thor's external documentation was linked by its README but could not be retrieved in this audit. Such cases remain access limitations rather than findings of missing documentation.

Each extracted claim has a source locator, evidence type, and qualification in S1. The record distinguishes an inspected artifact, an uninspected linked artifact, a specification, and a reported measurement. “Not established” means that the inspected material did not substantiate the stated property. It does not mean that the property is absent from the project or that another source could not establish it.

Extraction and subsequent checks followed one workflow; no independent second coding or inter-rater reliability assessment was performed. No arm was fabricated, assembled, calibrated, or physically tested. Firmware and control stacks were not installed. The resulting observations concern documentary evidence and its implications, not measured construction difficulty or hardware quality.

2.3 Comparability and source boundaries

Numerical values were retained only with their source meaning. An arm joint count was separated from gripper actuation where the documentation allowed that distinction. Height, maximum geometric reach, and a load-test offset were not combined. Costs were not converted into current equivalent prices because currencies, dates, component coverage, shipping, fabrication, and labor differed. Hardware-version changes were not assumed to preserve earlier performance.

A paper license applies to the article, whereas a design or software license applies to the corresponding artifact. The comparison records the scope stated in primary sources and identifies exceptions. It is a documentary analysis rather than a legal determination. Similarly, the public ISO 9283 catalog establishes the standard's subject, but its paywalled clauses were not inspected [10]. The proposed test package below is therefore not presented as ISO compliance.

3 A framework connecting releases to reproducible experiments

3.1 Seven domains and their dependencies

The proposed framework follows seven domains: identity and rights; design and fabrication; procurement; assembly and integration; commissioning; validation; and maintenance and change. Figure 1 shows their relationships. These domains extend established open-hardware documentation concerns into the configuration and test information needed for desktop manipulation [1]. They are questions to answer, not stages every project must publish in an identical format.

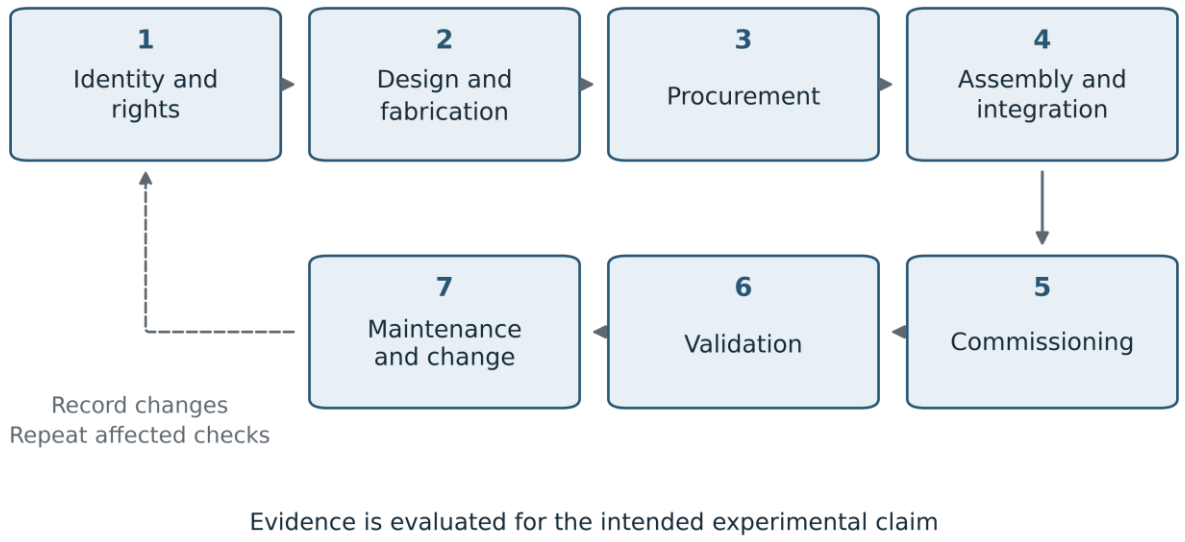


Figure 1. Seven domains connecting a public arm release to a specified experiment. Arrows organize the audit; they do not represent measured probabilities or a validated maturity model. A change may require renewed identity, calibration, or validation.

Identity and rights establish which release is being discussed and what reuse it permits. A project name alone is insufficient when an archived arm, a community revision, and a current kit share a lineage. A useful identity record links hardware files, firmware, controller configuration, and any derivative changes. Rights must be attached to the relevant files and dependencies. A license badge cannot resolve a specific exception elsewhere in the documentation.

Design and fabrication establish what can be made and modified. Native parametric files preserve design intent, constraints, and dimensions differently from neutral solids or tessellated meshes. A

STEP model can support geometric modification without preserving a native feature history. An STL mesh can support printing while leaving tolerance changes cumbersome. Consequently, these formats provide different affordances rather than interchangeable evidence of editable source.

Procurement connects the design to realizable components. A BOM should distinguish functional requirements from a single supplier link. A bearing identifier, shaft dimension, motor voltage, gear ratio, or connector pinout can remain useful after a shop page disappears. Substitution rules are especially important when visually similar components differ electrically or mechanically. The framework does not require identical vendors when an equivalent component is explicitly specified.

Assembly and integration explain how parts become a working mechanism. The relevant information includes order, orientation, fastening, cable routing, alignment, and transmission adjustment. Commissioning then establishes the relationship between commands, sensed states, and physical motion. Motor identifiers, direction conventions, joint zeros, limits, and calibration results belong to this domain. A successful connection to a motor bus is evidence of communication, not a geometric calibration.

Validation links the assembled configuration to a measured outcome. A payload, endpoint error, task success rate, or compliance response requires a procedure and boundary conditions. Maintenance and change preserve that relationship after re-tensioning, replacement, firmware updates, or sensor relocation. A modification can invalidate a calibration or comparison without changing the platform's familiar name.

The domains are interdependent for a specified claim. Reproducing a controller response requires configuration and test evidence, even when the printed structure is well documented. Extra detail in one domain cannot establish an unrelated missing condition. However, the framework does not assign equal weights or calculate an overall score. The relative importance of a missing item depends on the intended experiment.

3.2 Distinguishing artifacts, reconstruction, and replication

Artifact availability means that specified materials can be located and inspected. Reconstruction means that another build reaches a documented operational state. Experimental replication requires an appropriate repeat of the scientific test and comparison of its outcomes. These terms are used operationally here because reproducibility terminology varies across communities [2, 3].

Evidence supporting one level does not automatically establish the next. A complete BOM is not a successful build report. A demonstration video is not a position-repeatability distribution. A peer-reviewed learning result does not establish that all fabrication sources are released. Conversely, a project without a formal paper can provide strong assembly evidence. This separation prevents publication prestige or repository popularity from becoming a surrogate for the evidence being assessed.

For each performance statement, a compact record should connect the artifact version, physical configuration, procedure, result, and scope of inference. That record makes comparisons inspectable without requiring a single universal benchmark. It also supports provenance and reuse principles consistent with FAIR data stewardship [11]. FAIR principles motivate traceability here; they do not provide a certification of hardware reproducibility.

4 Comparative synthesis of the eight project families

4.1 Mechanical architecture changes the unresolved questions

Table 2 groups the cases by their principal actuation and construction arrangements. These families describe dominant dependencies rather than mutually exclusive robot categories. Printed structures, purchased actuators, and software interfaces recur across them.

Table 2. Mechanical families and the conditions they make important. The final column gives this review's engineering interpretation, not measured rankings.

Dominant arrangement	Cases and primary sources	Construction dependency	Evidence needed for a specified task
Smart-servo chain	OpenMANIPULATOR-X; Poppy Ergo Jr; SO-101 [12, 13, 14]	Purchased geared modules, horns, bus settings	Module variant, joint conventions, limits, tool and camera transforms
Stepper with reduction	Thor; Moveo; PAROL6 [15, 16, 17]	Belt or gear alignment, reduction, homing	Approach direction, backlash, load, current limits, thermal condition
Antagonistic tendons	GummiArm [4, 18]	Tendon geometry, tension, material and controller configuration	Joint calibration plus declared stiffness or co-contraction and material state
Hybrid actuation	Niryo One [19]	Stepper and servo configuration with linked hardware and software revisions	Hardware archive identity, motor configuration and matched controller state

Smart-servo chains place actuation, gearing, sensing, and communication inside purchased modules. OpenMANIPULATOR-X, Poppy Ergo Jr, and SO-101 illustrate this arrangement [12, 13, 14]. It reduces the number of separately designed transmission interfaces exposed to the builder. It also transfers important dependencies into actuator firmware, gear variants, bus configuration, and horn alignment. Encoder feedback inside an actuator does not by itself measure tool-center position or structural deformation.

Stepper-driven architectures expose more of the reduction and alignment chain. Thor uses printed gears, pulleys, and belts, while Moveo provides a printed educational mechanism with Marlin-based firmware [15, 16]. PAROL6 documentation describes joint-specific combinations of belts and planetary reduction [17]. For these arrangements, step commands and reduction ratios specify commanded motion. They do not measure backlash, elastic deflection, skipped steps, or endpoint repeatability.

GummiArm provides a different dependency through antagonistic tendon actuation [4]. Here, tension and material behavior are part of the operating configuration. The commissioning instructions explicitly account for assembly variation and tendon length [18]. The relevant inference is architectural: reproducing geometry does not establish equivalent force-displacement behavior. A configuration record for contact experiments therefore needs more than joint-zero values.

Niryo One is retained as a hybrid and licensing boundary case. Its hardware repository distributes printable parts and identifies an older archive for robots received before September 2018 [20]. The separate ROS stack contains stepper and Dynamixel control packages alongside the operational software [19]. These releases demonstrate why a project should be audited as linked artifacts. A hardware file archive alone cannot establish a complete reconstruction route.

4.2 Design files and procurement form different kinds of evidence

Table 3 summarizes the inspected artifact routes. The entries deliberately separate inspected files from external links. They should not be read as a ranking of the platforms.

Table 3. Documentary routes inspected for the eight project families. I = content or file inventory inspected; L = linked source not inspected successfully. An inventory observation does not validate CAD files or build completeness.

Project	Design and procurement	Assembly and commissioning	Limit of the inspected evidence
GummiArm	I: legacy native CAD inventory; selected CE solids/meshes; legacy build information [21, 22]	I: build and configuration wiki [18]	CE and legacy compatibility needs mapping; full cloud CAD L.
Thor	I: FreeCAD, STEP, STL inventories; BOM L [15]	Assembly and firmware guides L	External documentation access failed; README specifications inspected.
Moveo	I: CAD/STL inventories and BOM PDF [16]	I: assembly and firmware sections, manual pp. 6-45	Files and procedures inspected; reconstruction and quantitative test outcome not established.
OpenMANIPULATOR-X	I: parts list; CAD links L [12]	I: assembly page; detailed manual L; software repository I [23]	Specification and artifact links do not verify a modified build.
Niryo One	I: STL inventory and revision note [20]	I: ROS setup README; full self-build route not established [19]	Printable hardware repository is not a complete BOM or assembly package.
Poppy Ergo Jr	I: STEP/STL inventories, BOM and suppliers [13]	I: assembly-image inventory; external guide L; setup README I	Cloud-native design and complete external assembly guide not inspected.
PAROL6	I: STL inventory, limited STEP plates, BOM, print table [24]	I: illustrated manual and getting-started text [17]	Manual excludes complete STEP design and control-board openness.
SO-101	I: STEP/STL inventories and role-specific BOM [14]	I: assembly, addressing and calibration text [25]	Physical reconstruction and cross-copy performance not established.

The cases reveal several distinct meanings of a public hardware release. Thor exposes native FreeCAD, STEP, and printable files [15]. Moveo includes CAD, meshes, a BOM, and a combined user and assembly manual [16]. Poppy Ergo Jr provides STEP and STL files alongside its material list [13]. These are different but identifiable routes for fabrication and modification.

GummiArm also illustrates a version-mapping problem. Its legacy repository and community printables should not be interpreted as identical complete releases [21, 22]. The community README describes selected STEP and STL files and links to a fuller cloud-hosted design. That link was not tested as an editable model in this audit. Combining files across the lineage therefore requires an explicit compatibility decision.

A procurement list is most informative when it constrains functional equivalence. SO-101 documents different motor gearing for leader and follower roles and notes power-supply differences between servo variants [14, 25]. This information affects backdrivability and operation, not simply purchase price. It also shows why “six servos” is an inadequate substitute for a configuration-specific BOM.

Poppy Ergo Jr offers a published example of a sourcing-driven system change. Caravati and Tatar describe replacing the original Raspberry Pi and Pixl arrangement with an OpenCM9.04 board

because of import constraints [26]. The resulting platform remains recognizable as Ergo Jr, but its control pathway differs. That example supports recording substitutions as experimental configuration changes rather than treating procurement as a purely administrative stage.

The present comparison does not estimate which architecture minimizes total cost. A useful cost report would separate purchased parts, fabrication, failed prints, tools, assembly labor, calibration effort, and recurring maintenance. Their importance follows from the build process; their relative magnitudes were not measured here. Reported historical part costs should therefore remain dated source claims rather than current procurement promises.

4.3 Assembly instructions and commissioning serve different purposes

Moveo's manual separates mechanical assembly from firmware installation and use [16]. PAROL6 supplies an extensive illustrated assembly manual, a printing table, and separate operational documentation [24, 17]. These examples show that assembly evidence can be substantial even when other release domains remain restricted or incompletely verified.

Commissioning requires a further correspondence between mechanism and controller. The SO-101 instructions distinguish motor-address setup from calibration of leader and follower ranges [25]. The distinction is technically consequential: unique addresses allow communication, while calibration assigns a consistent interpretation to reported positions. Successful teleoperation after setup still does not quantify the residual error between two arms.

Legacy software instructions also require version-aware reading. The inspected GummiArm installation guide assumes ROS Kinetic, and the Niryo One ROS README specifies Ubuntu 16.04 and ROS Kinetic [18, 19]. The current OpenMANIPULATOR repository contains ROS 2 packages [23]. These observations do not establish that an older stack is unusable or that a newer stack builds successfully. They establish the need to distinguish a historical environment from a tested current environment.

For reconstruction, a useful endpoint is a recorded commissioning state. It should include joint directions, offsets, limits, controller settings, firmware identifiers, and a short motion check. For scientific reuse, that state should also preserve any task-specific calibration. Camera extrinsics, tool transforms, tendon preconditioning, or gripper changes may be essential depending on the experiment. A successful installation tutorial cannot establish all these relationships by itself.

4.4 Licensing must be inspected at the artifact level

Table 4 records the inspected licensing statements and their scope. It reports source text rather than declaring legal clearance.

Table 4. Licensing scope observed in primary artifacts. These entries are documentary findings, not legal advice or certification. Exceptions are retained rather than inferred away.

Project	Inspected statement	Scope or unresolved qualification
GummiArm lineage	Legacy LICENSE: CC BY-SA 4.0. CE README: CC BY-SA 4.0 [21, 22]	CE docs and printables LICENSE files name BY-SA but link to BY-NC-SA. The inconsistency is unresolved; do not infer unrestricted CE permissions.
Thor	Root CC BY-SA 4.0 [15]	Related controller, PCB, and ROS repositories require separate scope checks.
Moveo	Root MIT; Marlin_main.cpp header GPL v3 or later [16]	Root label does not replace retained firmware terms.

Project	Inspected statement	Scope or unresolved qualification
OpenMANIPULATOR-X	Software repository: Apache 2.0 [23]	Hardware CAD links and purchased modules were not assigned that software license.
Niryo One	Printable files: CC BY-NC-SA 4.0; ROS stack: GPLv3 [20, 19]	Noncommercial hardware terms differ from OSHWA's open-hardware definition.
Poppy Ergo Jr	Hardware: CC BY-SA 4.0; software: GPLv3 [13, 27]	Hardware and software scopes are explicitly separated.
PAROL6	Manual p. 4: GPLv3 for STL and stated software [24]	Complete STEP files and control board explicitly not open source in manual; limited STEP mounting plates are separately present.
SO-101	Repository root Apache 2.0 [14]	This observation is limited to the inspected release; component internals and third-party files need their own terms.

Niryo One's printable hardware uses a noncommercial Creative Commons license, whereas its separate ROS stack uses GPLv3 [20, 19]. OSHWA explicitly distinguishes noncommercial restrictions from open-source hardware [28]. Accordingly, Niryo One remains a useful documented comparison case without being classified here as unrestricted open hardware.

PAROL6 exposes another boundary. Its assembly manual states that the command software, STL files, and control-board software use GPLv3, while the full STEP files and control board are not open source [24]. The repository also contains a limited number of STEP mounting-plate files. Their presence does not overturn the manual's narrower statement about the complete robot design. Conversely, the manual's statement should not be used to deny the existence of those individual files.

Moveo's root license is MIT, while the inspected Marlin source header retains GPL terms [16]. Poppy explicitly separates a Creative Commons hardware license from GPL software [13, 27]. These examples show why a single repository badge is an incomplete description of a multi-artifact release.

The GummiArm community files also contain an unresolved inconsistency. Their README statements identify CC BY-SA 4.0, but the short LICENSE files link to BY-NC-SA while naming BY-SA [22, 18]. The legacy repository's license link is internally consistent [21]. This audit cannot determine which community-edition statement expresses the intended grant.

An appropriate release manifest should identify the license for mechanical sources, electronics, firmware, host software, documentation, and contributed files. Purchased modules require their own documentation and terms. The present review neither infers rights to proprietary actuator internals nor treats a permissive software license as evidence that electronics design files are available.

4.5 Performance evidence needs conditions before comparison

Table 5 contrasts several forms of performance evidence. It intentionally avoids a cross-project leaderboard.

Table 5. Examples of performance evidence and its interpretation. No common test was performed and no ranking is implied.

Source and metric	Published value or description	Evidence class	Permissible interpretation
GummiArm v1.0 load capacity [4]	0.75, 0.65, 0.45 kg at forward offsets 300, 405, 510 mm	Reported measurement	Pose- and version-specific capacities; no uncertainty or trial count accompanies these three values in the discussion.
OpenMANIPULATOR-X [12]	500 g payload; 380 mm reach; repeatability <0.2 mm	Manufacturer specification	Source specifications; not an independently verified modified-build result.
Thor [15]	625 mm stretched height; 750 g maximum payload including end effector	Project specification	Height is not a horizontal reach test; load criterion and repeated endpoint test not established here.
PAROL6 [17]	Joint-level theoretical microstep sizes; nominal 0.1 mm repeatability	Calculation and specification	Command resolution differs from measured endpoint repeatability; common test conditions not established here.
Learning-system papers [29, 30]	Task and teleoperation evaluations	System-level experimental report	Evidence concerns the reported full setup; it is not a mechanical precision comparison across these arms.

The OpenMANIPULATOR-X specification lists a 500 g payload, 380 mm reach, and repeatability below 0.2 mm [12]. These are manufacturer specifications in the inspected page. They are not measurements performed in this review, nor do they establish equivalent performance after modifying printed links or assembly. The page's count of five degrees of freedom explicitly includes a gripper, leaving four arm-motion joints.

Thor's README distinguishes a 625 mm stretched height from a maximum payload that includes the end effector [15]. Treating that height as a tested horizontal reach would change the meaning of the source. PAROL6 specifications similarly publish theoretical joint step sizes separately from system specifications [17]. Such resolution calculations are not measured endpoint accuracy.

The original GummiArm v1.0 report supplies a particularly useful conditional example [4]. Figure 2 replots its three stated load capacities against the reported forward offsets. The plot preserves the historical version and coordinate description. It supplies no new measurements, uncertainty estimates, or fitted load model.

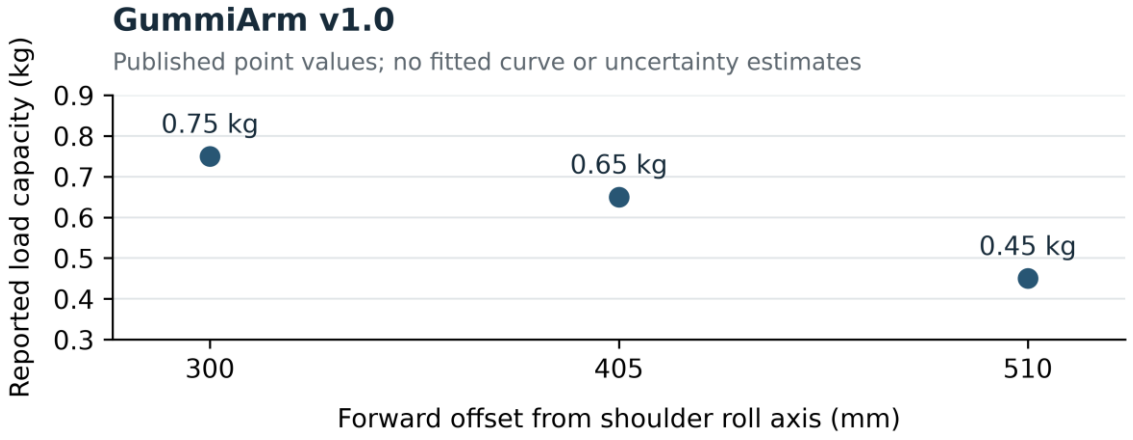


Figure 2. Source-reported GummiArm v1.0 capacities [4], p. 16. Forward offsets are from the shoulder roll axis. No uncertainty or trial count accompanies these values; no fitted curve or later-version transfer is implied.

The correct lesson is methodological. A load capacity requires a pose, configuration, and acceptance criterion. A single scalar cannot describe every operating condition. The review does not extrapolate these three observations into a workspace envelope or transfer them to the community-edition arm.

For position performance, accuracy and repeatability address different questions. A device may consistently revisit a location while remaining offset from its commanded pose. Comparisons need the target definition, approach direction, load, speed, tool transform, reference instrument, repetitions, and reported statistic. ISO 9283 provides an established industrial-robot performance context [10]. Any desktop test adapted from that context should report its adaptations and avoid unsupported claims of conformance.

Task success adds another layer. It combines the mechanism, controller, perception, object properties, initial conditions, and success rule. A learned policy's success rate therefore does not identify the arm's mechanical repeatability. Equally, a modest geometric specification does not establish that a device cannot perform a visually guided task. Keeping these outcomes separate is necessary for a fair interpretation of low-cost manipulation.

5 Implications for embodied intelligence and robot learning

5.1 Reproducing the observation and action interface

A research platform includes the interface through which a policy observes and acts. Relevant details include joint order, units, signs, normalization, command type, update timing, camera configuration, and gripper semantics. When those details change, a numerically identical action vector can describe a different physical command. This conclusion follows from the interface definition and does not assume that every mismatch causes a failed policy.

ALOHA demonstrates that learning and teleoperation can support demanding manipulation with comparatively inexpensive hardware [29]. Its experimental result concerns a complete system, including demonstration collection and a learning method. GELLO supplies low-cost kinematically corresponding input devices for several target arms [30]. Its input-device design does not make the driven industrial arm an open fabrication project. These distinctions prevent a teleoperation release from being mistaken for an unrestricted release of the entire manipulation system.

SO-101 and LeRobot connect an accessible arm to explicit setup and calibration procedures [14, 25]. However, a documented mapping between joint ranges is not evidence that every assembled copy shares the same dynamics. An experiment intended for transfer should preserve the interface schema and report physical deviations separately. Dataset and policy compatibility should be tested under the intended conditions rather than inferred from a common platform label.

5.2 Compliance and adaptation change the comparison

Mechanical compliance can be an intended experimental variable. Elstner and colleagues study contact-rich learning with GH360, a developed GummiArm descendant, using exploration and demonstrations [31]. Its changed actuation and sensing distinguish that setup from the earlier public arms. This supplies contemporary evidence of the lineage's research use. It does not independently verify the buildability of every public release or establish certified safety for other applications.

Li and colleagues demonstrate visuomotor control across diverse systems, including a manually assembled low-cost arm with substantial backlash [32]. Their result supports distinguishing task-level adaptation from nominal geometric precision. It does not show that manufacturing tolerances or calibration cease to matter. Rather, the experiment's observations, training conditions, and controller become part of the evidence required to reproduce its performance.

For compliant arms, a benchmark should state whether stiffness is fixed, commanded, estimated, or changed during a trial. For geared arms, it should state whether the approach direction reverses. For visually guided tasks, camera placement and latency belong to the configuration. These are architecture-sensitive reporting choices, not competing definitions of a universally best arm.

ROBEL provides a useful external comparator because it reports replication across robot copies at different institutions and supplies benchmark materials [33]. Its hand and quadruped are outside the core desktop-arm sample. The transferable lesson concerns reporting an actual replication outcome instead of equating an available release with a replicated result.

6 A minimum release package and verification route

Table 6 translates the synthesis into a proposed minimum release package. The package is intended for research releases and experiment supplements, not as a retrospective pass-fail certification of community projects.

Table 6. Proposed minimum release package. This is a recommendation derived from the synthesis, not a validated certification checklist.

Domain	Minimum inspectable record	Question resolved
Identity and rights	Build ID; immutable source versions; per-artifact licenses and exceptions	Which release, and which reuse rights?
Design and fabrication	Native/neutral/mesh roles; dimensions; material; print orientation; tolerances where relevant	What can be fabricated and modified?
Procurement	Quantities; part identifiers; voltage, gearing, dimensions; permitted substitutes; dated cost coverage	Which substitutions preserve the design?
Assembly and integration	Sequence; alignment; wiring and pinouts; fastening and transmission adjustment	How are parts integrated consistently?
Commissioning	Joint signs, zeros, limits; firmware and controller settings; calibration procedure and output	What do commands and observations mean?
Validation	Load and pose; motion and sensor conditions; repetitions; raw data; analysis; failure criteria	What was measured, and under which conditions?
Maintenance and change	Replacements; wear or tension state; recalibration triggers; retest and failure logs	When does an earlier comparison stop applying?

A build manifest should begin with immutable identifiers for mechanical files, electronics, firmware, software, and documentation. It should connect these identifiers to the physical serial or local build identifier and list substitutions. A compatibility statement can be more useful than a requirement for identical vendors. The statement should specify which properties must be preserved and which changes require renewed calibration.

The procurement and fabrication records should identify dimensions, tolerances, materials, print orientation, and component variants where they affect fit or behavior. Assembly instructions should identify alignment and adjustment points explicitly. Commissioning outputs should be machine-readable where feasible, with joint conventions and calibration procedures explained in ordinary text. A saved configuration file without its physical meaning remains difficult to reuse.

Verification should proceed from controlled basic checks to the intended task. Communication and direction checks establish that the correct joints respond. A joint-zero and limit check establishes a configuration baseline. A load-at-pose check examines a stated operating condition. A position-return

test characterizes repeated motion under a declared approach and load. A timing trace records command, state, and sensor relationships when learned control depends on them.

These are proposed test categories, not experiments conducted here. The number of poses, trials, and acceptance thresholds should be selected for the target claim. A precision-placement study needs stronger geometric evidence than a qualitative teaching demonstration. A contact-learning study may need force-displacement or stiffness characterization that would add little to a simple drawing task. The protocol should justify these choices before interpreting the results.

Each test should release the raw records, analysis script, and result summary together with the configuration identifier. When a baseline fails, that failure should remain in the record alongside the repair and repeat test. This makes maintenance scientifically interpretable. Replacing a belt or changing a camera mount can then be traced to the experiment it affects.

The proposed package also includes safe operating boundaries and stop behavior. A license disclaimer is not a tested safety function. Reported compliant behavior is not certification for human contact. For learning experiments, descriptions should identify how exploration is constrained and how interrupted trials are recorded. No specific safety integrity level or collaborative-robot classification is inferred for the reviewed arms.

7 Discussion and research priorities

7.1 What the comparison establishes

The inspected cases support a specific conclusion: artifact openness and experimentally demonstrated reproducibility should be reported separately. Several releases provide substantial construction information, but that information has different scopes. Licensing exceptions, linked-only CAD, historical software environments, and condition-dependent specifications each require a distinct interpretation. A single completeness score would obscure those differences without evidence that its weights predict build or research outcomes.

The proposed framework adds resolution to earlier open-hardware guidance by connecting release domains to experimental claims. It does not replace broad surveys, hardware metapapers, or industrial performance standards. The contribution is a comparison structure for identifying the dependency that remains unresolved for a particular task. Its usefulness as a selection tool still requires evaluation with actual builders and independent replications.

The architecture comparison also suggests that documentation effort should follow the uncertainty that matters. A smart-servo chain concentrates dependencies in module variants and configuration. A printed reduction mechanism exposes alignment and transmission adjustment. An antagonistic tendon mechanism makes tension and material state part of the configuration. These are engineering interpretations of the inspected designs, not measured differences in construction success or maintenance cost.

7.2 Limits of the evidence

The purposive sample supports mechanism-based comparison, not prevalence estimates for all desktop arms. Search-engine coverage and ranking can miss projects, languages, or unindexed documentation. Official repositories favor well-maintained visible releases, while successful informal reconstructions may remain undocumented. Adding more cases could change the range of release patterns without invalidating the need to distinguish their evidential roles.

The audit is also bounded by accessibility. Some links failed, cloud CAD was not opened, and videos were not reviewed as complete assembly procedures. Source inventories were inspected without loading the CAD designs or executing the controllers. An inaccessible document therefore contributes uncertainty, not proof of inadequacy. The supplementary evidence records preserve this distinction so that later readers can update the comparison.

No independent extraction, physical build, or current procurement exercise was performed. Consequently, the review cannot rank true assembly difficulty, durability, total cost, or performance. Documentary coverage is not a substitute for those outcomes. Primary reports may also omit test details or emphasize successful demonstrations. The review preserves those limitations rather than filling them with estimated values.

Repository commits improve traceability but do not freeze external vendors, hosted documentation, firmware downloads, or cloud services. A durable archive of the review's own extraction files is therefore valuable, while third-party redistribution must respect the applicable terms. The supplied source records and checksums support audit of this review; they are not a promise that every dependency will remain obtainable.

7.3 The next informative studies

Two studies would test the practical value of this synthesis. First, independent teams could reconstruct the same frozen release and record substitutions, assembly interventions, commissioning outcomes, and elapsed effort. Comparing their logs would reveal whether the proposed domains capture the decisions that actually block reconstruction. Unanticipated failure categories would require revising the framework.

Second, a shared experiment could compare independently assembled copies before and after a declared calibration procedure. The comparison should separate geometric error, dynamic or contact behavior, and task success. This would test whether matching a release and interface sufficiently constrains the physical system for the intended claim. It would also identify when adaptive control compensates for differences and when it hides a change in experimental conditions.

Both studies should publish unsuccessful attempts and maintenance interventions. Without those records, a release can appear straightforward because undocumented expert work disappears from the account. Evidence of independent reconstruction would provide a stronger basis for future platform comparisons than additional nominal specifications alone.

8 Conclusion

Open desktop robot arms are useful research resources when their releases make the relevant construction and experimental conditions inspectable. The eight project families show why mechanical sources, BOMs, assembly documents, licenses, commissioning records, and performance reports answer different questions. A seven-domain framework connects those questions without converting documentary coverage into an unvalidated score. The proposed release package links a physical build to calibration, test conditions, and subsequent changes. For embodied intelligence, it also preserves the observation and action interface through which behavior is learned and evaluated. The next step is independent reconstruction and shared testing of frozen releases. Until that evidence is available, source availability should support a bounded claim about accessible artifacts, with broader claims tied to the experiments that establish them.

Data availability statement

No new robot experiments were performed. The extracted evidence records, search log, project profiles, repository identifiers, numerical figure source data, and analysis and figure-generation scripts are supplied in Supporting Information S1. Third-party papers and project files remain available from their cited sources under their respective terms. Locally retained source copies support verification and are not part of the redistribution package.

Ethics statement

This review used public technical literature and project documentation. It involved no new human-participant or animal research and no collection of personal participant data.

Supporting information

S1 contains the search record, claim-level evidence table, project profiles, source register, repository snapshots, numerical figure source data, and reproducibility scripts. Its README explains evidence-status codes, access limitations, and the scope of the documentary audit.

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